

# Adding a reusable representation beside a fixed prediction service: an evaluated problem, a coalition-conditioning effect, and two negative method results

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## Abstract

An organisation already publishes a prediction service: fixed probability vectors that downstream recipients consume and cannot be asked to re-accept. We ask what happens when it adds a *second*, reusable feature channel for one recipient without changing a single published number, and tries to limit what that recipient—and a coalition of recipients—can *newly* infer about sex and race. We contribute an evaluated formulation of that problem and two completed studies of one mechanism family for it. The mechanism keeps the teacher information the published outputs do not explain and chooses directions by maximising linear reconstruction of that residual minus conditional-moment penalties, conditioning on the released service probabilities through cross-fitted nuisances. Penalties can be local to one recipient or specific to a coalition’s joint view. Five findings. (i) Exact service preservation is structural and holds bitwise on all 210 released views; it does not imply stable service *accuracy*, which moves by up to 0.015 nats across one year. (ii) On a locked, previously unused survey year the coalition-conditioned penalty beats both an equal-strength and an equal-total-mass local control under simultaneous 95% intervals: 14 of its 16 sensitive endpoint cells are strictly better, none is worse, and two are unresolved. The registered rule—at least one strictly better and none strictly worse, per weighting—passed in all four cells. (iii) That rule pairs with a *point* constraint on residence utility, and the interval evidence does not establish non-inferiority at the same 0.001 margin: zero of four comparisons, under any procedure we computed. (iv) Under identical attacks the mechanism is significantly worse than a frozen adversarially trained channel of the same width on all four sensitive endpoints under both weightings, and that ordering survives matching the two on average utility in both directions. (v) A refinement—a penalty nonlinear in the channel, plus a rank-8 compression—moves measured recovery substantially in the intended direction on development data, but all eight registered coordination cells still fail, the best candidate remains significantly worse than the neural channel on local sex, and an exploratory cross-year look widens rather than narrows that gap. The refinement also turned out to be coordinate-dependent: a search that only rotates the released subspace, changing nothing an attacker can recover, reaches a large share of its training-objective gain. We report an evaluation contribution and two negative method results, and we list every claim of our own that this revision withdraws or narrows.

## 1 What this paper is, and what it is not

A service that already publishes predictions can hand one recipient extra features built only from what those predictions leave unexplained, and can steer those features away from directions that would let that recipient—or that recipient plus a colluding one—guess sex or race better than the published predictions already allow. We show the steering knob works on a sealed new year of data against its own local controls; that a refinement of it moves recovery further in the intended direction on development data; and that neither version is competitive with an adversarially trained channel of the same size.

**This is an empirical and problem-formulation paper, not a method paper.** The useful output is the formulation—fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles—together with the evaluation design and two honest negative results. It is not a deployable mechanism and we do not present it as one. No claim here establishes universal privacy, conditional independence, an upper bound on mutual information, or the failure of any mathematical method family.

Two completed studies are reported. **Study 1** is a locked temporal transport of a frozen 14-interface slate to a previously unused survey year (§7). **Study 2** is a development study of a nonlinear-moment refinement and a rank intervention (§8). The main text keeps them apart by what each is evidence *for*, not by the order in which they were run. §9 lists every claim this revision withdraws or narrows, with the evidence for each.

**One paragraph for a colleague.** We study a release setting that existing invariant- and fair-representation work does not evaluate: an organisation has already published prediction outputs it cannot revise, and wants to add a reusable feature channel for one recipient while limiting what that recipient, and that recipient colluding with another, can newly infer about protected attributes. Within that setting we show that a penalty built specifically from the coalition’s joint view measurably reduces recovery relative to local-only penalties on a sealed evaluation year—but that the whole mechanism family we tried remains clearly behind an adversarially trained channel of the same width, so what we are offering is the evaluated problem and the measurement design, not a competitive algorithm.

## 2 Threat model and interface, in plain language

Each person has covariates  $X$ , sensitive attributes  $S$  (sex, 2 classes; race, 9 classes) and task labels. Before anything in this paper happens, two services are already frozen and published:

- $H_A(X) \in [0, 1]^4$  — the full income-over-\$50k and employment probability vectors, released to recipient A;
- $H_B(X) \in [0, 1]^2$  — the full public-coverage probability vector, released to recipient B.

A mechanism may add a channel  $Z \in \mathbb{R}^{16}$  *for A only*. Recipient B’s release is unchanged. So the three views are: A sees  $(H_A, Z)$ ; B sees  $H_B$ ; the coalition AB sees  $(H_A, Z, H_B)$ . Throughout,  $H$  with no subscript means the no-channel interface: what an attacker can already do from the published service alone.

Three names recur and are fixed here once.

**$J$ , the frozen neural channel.** A 16-dimensional adversarially trained channel, recovered from the earlier line of work and never retrained here. It is the strongest comparator in the study. It is *internal*: it is not an external published baseline, and no result below should be read as a comparison against a state-of-the-art system (§5).

**Local controls,  $L1$  and  $L2$ .** Spectral arms whose penalty uses only recipient A’s own view.  $L1$  is matched to the candidate on penalty *strength*;  $L2$  is matched on total penalty *mass*. They exist so that a coalition effect cannot be confused with “more penalty”.

**Coalition condition,  $C1$ .** The same construction with an additional penalty term built from the coalition’s joint view  $(H_A, Z, H_B)$ . It is the candidate.

The adversary holds labelled data from the release year, fits fixed attack families on its own pool, selects on a validation pool, and sees only its legal view. The policy authorises A for income, employment and residence, B for coverage and commute, and forbids sex and race everywhere, coverage and commute for A, and income, employment and residence for B—eleven forbidden roles. Every published function, including the preprocessing, the released map and the neural arms’ public auxiliary heads, is in scope, and coalition attacks may use any legal projection of A’s and B’s candidates.

**The A-only limitation is real and bounds every claim below.** Only one recipient receives a channel. Nothing here demonstrates joint optimisation of channels for several recipients, and the coalition penalty is a constraint on one channel given another recipient’s fixed view, not a multi-channel design.

## 3 Why the published outputs are retained

The constraint that makes this problem different from representation learning is that  $H_A$  and  $H_B$  have already been published. Downstream consumers have built on those exact numbers; a revision is a migration,

not a deployment. So the mechanism may append but not modify:  $Z$  is concatenated to  $H_A$  and never mixed into it. Exact preservation is then *structural* rather than measured—there is no perturbation whose size must be bounded—and we verify it bitwise on every released view rather than certifying it statistically. This also means preservation is *easy*: appending a vector while leaving another untouched is a trivial observation about concatenation, not a theorem, and we claim nothing more for it.

Two things it does not give us. It does not mean the service stays *accurate*: on identical released vectors, employment log loss rises by 0.0138 nats and income falls by 0.0145 between the two years (Table 8). And it does not constrain what the appended channel discloses, which is the entire subject of this paper.

## 4 The mechanism

### 4.1 The original fixed-matrix construction

Let  $T$  be a frozen PCA32 teacher. Build  $\phi(T) = [\text{standardised } T, 96 \text{ random Fourier features}]$  with a median-heuristic bandwidth [Rahimi and Recht, 2007]. Let  $q_A$  be the intercept and all degree-2 monomials in  $A$ 's two positive-class service probabilities;  $q_{AB}$  adds  $B$ 's. Residualise by SVD least squares,  $V_0 = \phi - q_A B \phi$  and  $R = T - q_A B T$ , then eigen-whiten  $V_0$  to  $V$  with  $V^\top V/n = I$ . Utility is  $U = C_{VR} C_{RV}$ : for  $W^\top W = I$ ,  $\text{tr}(W^\top U W)$  is exactly the training reduction in least-squares reconstruction error of the residualised teacher. Protection matrices use household-grouped 3-fold cross-fitted multinomial-logistic nuisances  $m_j$  (cross-fitting in the sense of Chernozhukov et al., 2016; we claim none of that work's orthogonality or rate guarantees for our moments) and, for each retained basis column  $b_k$ ,

$$G_{jk} = \frac{1}{n_j} \sum_i V_i b_k(H_i) (\text{onehot}(S_j)_i - m_j(H_i))^\top, \quad P_j = \sum_k G_{jk} G_{jk}^\top \text{ (trace-normalised),}$$

averaged over local roles (basis  $q_A$ : sex, race, coverage), coalition roles (basis  $q_{AB}$ : sex, race) and a marginal variant (constant basis). The released map is the top-16 eigenvectors of  $U_{\text{norm}} - \lambda P$ , giving arms S0 ( $\lambda = 0$ ), M025/M1 (marginal), L025/L1/L2 (local) and C025/C1 (local + coalition). Inference needs only  $T$  and  $H_A$ ;  $H_B$  enters fitting and auditing only.

### 4.2 The refinement studied in Study 2

Study 2 replaces the fixed penalty matrix with a penalty computed from degree-two monomials of the channel output itself, so that the objective is no longer of the form  $\text{tr}(W^\top A W)$  for any  $W$ -independent  $A$ . It also adds a predeclared rank-8 variant. Both are development-only interventions. The refinement was fitted, audited and then found to be *coordinate-dependent*; §8.5 reports that diagnostic. **The defect was not repaired in the maps that were evaluated**, so every Study 2 number below is a number for the defective objective, and should be read that way.

### 4.3 What is a theorem here, and what is not

The eigenvector step is Ky Fan's maximum principle for a *fixed* empirical matrix [Fan, 1949]. It certifies nothing about the choice of features, basis, nuisances, rank or attacker, and with  $r$  fixed at 16 it can retain directions whose penalty exceeds their utility. Crucially, it is a statement about the *original* family only: the refinement of §4.2 has no fixed matrix to take a spectrum of, so it inherits no optimality statement whatever. Selecting rank by eigenvalue sign is not our idea and is not new: it is SARL's Theorem 3, OptNet-ARL's Theorem 4.1 and K-TOpt's Corollary 4.1 [Sadeghi et al., 2019, 2021, 2022]. We fixed  $r = 16$  to match the width of the neural baselines, which is a comparison choice, not an optimum.

The moment condition is a finite subset of the  $L^2$  characterisation of conditional independence that kernel CI tests use [Zhang et al., 2011, Strobl et al., 2019]. Precisely, it is Lemma 2(v) of Zhang et al. [2011], attributed there to Daudin: our centring on the  $S$  side is algebraically equivalent to residualising the  $(Z, H)$  side, since with  $\tilde{f} = f - \mathbb{E}[f | H]$  we have  $\mathbb{E}[f(Z, H)(\mathbf{1}\{S = c\} - m_c(H))] = \mathbb{E}[\tilde{f} \cdot \mathbf{1}\{S = c\}]$ . The characterisation quantifies over *every* square-integrable  $f$ ; we test  $f \in \{Z_\ell b_k(H)\}$ —linear in  $Z$ , degree-2 in

$H$ , with estimated nuisances and finite samples. Consequently vanishing fitted moments do not establish conditional independence, and a non-zero moment may reflect nuisance error rather than leakage. Two fixtures make this concrete and are carried in the code. An XOR channel has exactly zero marginal moment while  $S = ZH$  recovers it perfectly; it is detected by an interaction term *in*  $H$ , so enriching the conditioning side suffices. A magnitude channel— $(Z, S)$  uniform on  $\{(-1, 0), (1, 0), (-2, 1), (2, 1)\}$  with  $H$  constant—has exactly zero first moment, yet  $\mathbf{1}\{|Z| > 1.5\}$  recovers  $S$  exactly; because  $H$  is constant, *no* enrichment of the conditioning side can ever see it, and only a nonlinear function of  $Z$  can ( $\mathbb{E}[Z^2(S - \frac{1}{2})] = 0.75$ ). The attacks that defeat our arms on ACS are MLPs and boosted trees, i.e. the second kind.

With population-optimal log-loss predictors, the  $H$  versus  $(H, Z)$  risk difference equals  $I(S; Z \mid H)$ . Our validation-selected finite attacks do not attain those infima, so the reported differences bound it in neither direction—indeed some are negative—and mutual-information estimation has its own formal limits [McAllester and Stratos, 2020]. We report no certified information quantity. For the withholding control, an independent visible branch  $B \sim \text{Bernoulli}(p)$  gives  $I(S; B, Z_B \mid H) = pI(S; Z \mid H)$  and expected routed loss  $(1 - p)L_H + pL_{HZ}$ ; the branch must be persistent, since independent re-draws would eventually reveal  $Z$ .

**Invariance, stated carefully.** For an invertible coordinate change  $W \mapsto WQ$  with  $Q$  orthogonal, the released *information* and the unrestricted Bayes-optimal prediction risks are unchanged, because the two releases are measurable functions of one another. That is a statement about information and about optima over all measurable predictors. It is *not* a statement that every finite trained attacker, or every feature-restricted penalty, behaves identically on the two releases: a fitted MLP at a fixed budget is a particular algorithm applied to particular coordinates. We use the first statement only, and only where it is what the argument needs.

## 5 Relation to prior work

The closed-form line of Sadeghi, Dehdashtian and Boddeti already solves this shape of problem. SARL [Sadeghi et al., 2019] takes eigenvectors of a utility-minus- $\lambda$ -dependence matrix; OptNet-ARL [Sadeghi et al., 2021] keeps closed-form ridge players inside a learned encoder and already handles several sensitive attributes through several  $\lambda$ ; K-TOpt [Sadeghi et al., 2022] adds random Fourier features and evaluates on Folktables [Ding et al., 2021]; U-FaTE [Dehdashtian et al., 2024] adds a *conditional* penalty. Our marginal arms are, up to parametrisation, SARL’s construction on whitened random-feature residuals. **Reusing a known solver, a conditional-moment characterisation or a rank rule is attribution, not innovation**, and we list the reused components explicitly in `RELATED_WORK_SCOPE.md`.

Three things differ, and we state their weight honestly. *What we condition on*: U-FaTE’s conditional measure is stratification over a discrete label  $Y$ , and its own text scopes the equivalence to “when  $Y$  is not a continuous label”;  $H_A$  is a pair of continuous probability vectors, so stratification is unavailable and we substitute cross-fitted nuisance residuals.  $H$  is also a *released artifact* rather than the label to be predicted—a different object even setting continuity aside. *What utility means*: reconstruction of a residualised teacher rather than dependence with a label. *Recipient structure*: separate local and coalition penalties, with only one recipient receiving a channel.

The first two are adaptations of known components, not new optimisation principles. The third we did not find in the surveyed literature, and the nearest candidate is weaker prior art than its title suggests: the multi-consumer information-theoretic line [Sankar et al., 2013] designs *one* sanitised release consumed by many users who differ only in private side information, with no per-recipient release, no collusion term, and no constraint to preserve an already-published output. Linear concept erasure [Belrose et al., 2023, Holstege et al., 2025] addresses a different guarantee (exact *linear* erasure) on a representation the releaser owns and may edit; LEACE scopes its guarantee to linear adversaries explicitly. Adversarial removal is known to leave recoverable information [Elazar and Goldberg, 2018], and least-privilege learning has fundamental limits [Stadler et al., 2024].

Two further neighbourhoods are worth separating from ours. Conditional fair representation learning conditions on a *label* for separation-style criteria [Zhao et al., 2019, Hwa et al., 2024], not on a released prediction, and again designs the representation from scratch. Information-theoretic privacy gives the

| Data pool                | Role                      | Selection saw it?      | Intervals?             | Used by         |
|--------------------------|---------------------------|------------------------|------------------------|-----------------|
| ACS 2018                 | development               | yes, repeatedly        | yes (dev. bootstrap)   | both studies    |
| ACS 2017, original seal  | <b>confirmatory</b>       | no                     | yes, simultaneous      | transport study |
| ACS 2017, after the seal | exploratory reuse         | 2017 fitting/val. only | <b>none, by design</b> | nonlinear study |
| ACS 2016                 | admitted, <b>unscored</b> | no                     | n/a                    | nothing yet     |

**Table 1:** The four evidence roles, kept distinct in the abstract, the figures, the methods and the limitations. The original 2017 seal is the only confirmatory evidence in this paper. Its later reuse by Study 2 is development evidence on an exposed partition and carries no intervals by design. ACS 2016 is admitted on documented provenance and schema checks with a prospective label-blind household split prepared, and is **unscored**: no model was fitted on it, no outcome distribution was inspected, and no performance diagnostic was produced.

quantities our log-loss endpoints relate to—leakage and utility both become mutual information under log loss [Makhdoumi et al., 2014], with perfect-privacy conditions [Rassouli and Gündüz, 2021] and robustness of leakage measures to side information [Liao et al., 2019]—but assumes a known joint distribution and a randomised mapping, whereas our released map is deterministic and our attacks are fitted. Task-agnostic private representation learning [Arevalo et al., 2024] targets attribute inference without a known downstream task, again with no fixed prior output and one recipient.

**The scope of “we did not find it”.** A bounded search cannot establish novelty and we make no priority claim. We distinguish four questions and answer them separately in `RELATED_WORK_SCOPE.md`: whether a prior formulation *can express* this setting (several can); whether it supplies an *algorithm* applicable here (several do, with adaptation); whether it *evaluated* recipient-specific and combined-access policies with unchanged existing outputs (we did not find one that did); and whether our study demonstrates an effect beyond its own controls (§7.2: yes, against  $L1$  and  $L2$  on one sealed year). “We did not find a directly matching evaluation” is the claim; “no prior method can do this” is not, and would be false for at least the adaptation question.

**No external baseline was run, and what that costs us.** SARL, K-TOpt, U-FaTE, OptNet-ARL, LEACE and SPLINCE are discussed but never executed here. So this paper cannot claim competitiveness with any of them and does not. In particular,  $J$  is an *internal* adversarially trained channel, not an external benchmark, and where  $J$  beats our arms that is evidence about  $J$ , not about the state of the art. `RELATED_WORK_SCOPE.md` specifies what a fair external comparison would require, including two constraints that are easy to get wrong: if  $H$  must remain immutable then LEACE/SPLINCE may act only on the auxiliary channel  $Z$ , and any guarantee they carry then has that scope; and a baseline may not consume residence labels if it is to be compared on transfer to a residence task excluded from representation fitting. Any privileged-label variant is an oracle and belongs outside the main comparison.

## 6 Experimental design and the status of each data pool

California ACS one-year PUMS, ages 19–34, three household splits (seeds) throughout, with disjoint pools for representation fitting, source validation, task fitting and validation, attacker fitting and validation, and evaluation.

**Study 1: locked transport.** Fourteen interfaces per seed—H, E, A0, L025, L20, J (frozen adversarially trained channels, recovered without retraining) and the eight spectral arms—i.e. 42 interface instances, of which 24 are the new maps, frozen before any reserved-label evaluation. Development was ACS 2018. Transport was ACS 2017, admitted on provenance and schema alone, with a label-blind household hash partition fixed before any prediction existed. **Mode A** applies everything frozen: 2018 services, channels, probes, attacker candidate sets and 2018 validation selections are scored once on 2017. **Mode B** refits only probes, priors and attackers on 2017 fitting pools and selects on 2017 validation pools; representations,

| Interface      | Objective                              | $r$ | Fitting labels    | Inf. inputs | Role                |
|----------------|----------------------------------------|-----|-------------------|-------------|---------------------|
| H              | none (no channel)                      | —   | none              | $H_A$       | reference           |
| E              | adversarial, frozen                    | 16  | sex, race         | $X$         | context             |
| A0             | adversarial, frozen                    | 16  | sex, race         | $X$         | context             |
| L025           | adversarial, frozen                    | 16  | sex, race         | $X$         | F2 control          |
| L20            | adversarial, frozen                    | 16  | sex, race         | $X$         | F2 control          |
| J              | adversarial, frozen                    | 16  | sex, race         | $X$         | neural ref.         |
| S0             | $U$ only ( $\lambda = 0$ )             | 16  | none              | $T, H_A$    | context             |
| M025 / M1      | $U - \lambda P_{\text{marg}}$          | 16  | sex, race         | $T, H_A$    | context             |
| L1 / L2 / L025 | $U - \lambda P_{\text{loc}}$           | 16  | sex, race, cov.   | $T, H_A$    | F1/F4 control       |
| C025 / C1      | $U - \lambda(P_{\text{loc}} + P_{AB})$ | 16  | + coal. sex, race | $T, H_A$    | candidate           |
| lin16-*        | $U - \lambda P$ (alias)                | 16  | as above          | $T, H_A$    | alias, not refitted |
| n1r16-*        | nonlinear refinement                   | 16  | as above          | $T, H_A$    | dev. only           |
| lin8-*         | $U - \lambda P$ , compressed           | 8   | as above          | $T, H_A$    | dev. only           |
| n1r8-*         | nonlinear ref., compressed             | 8   | as above          | $T, H_A$    | dev. only           |

**Table 2:** The condition table. Every interface is fitted on ACS 2018 source data. “Fitting labels” are what the representation’s penalty consumed; “inference inputs” are what the released map needs at serving time. **Residence and commute labels are absent from every representation fit.**  $H_B$  enters fitting and auditing for coalition arms and is never an inference input. The lin16, n1r\* and lin8 rows are Study 2 only. Fit counts and the full grid are in Appendix C.

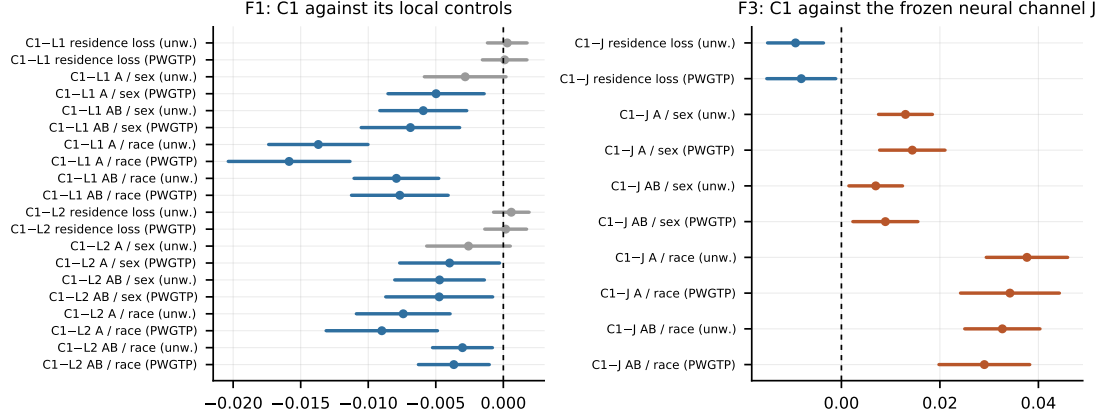
protection matrices and nuisances are never refitted. The two are never pooled: a weaker frozen attack after a shift is not evidence that fresh attackers fail.

Attacks are logistic regression, two restarted MLPs at 120/360 epochs, two gradient-boosted trees and a random-feature kernel-ridge family, plus every legal projection of H’s own attacks and of singleton candidates into the coalition view, plus, for the neural arms, their public auxiliary heads and saved-observer catch-up trajectories. Before the final partition could be read, a lock hashed 17,639 inputs—protocol, comparison families, code, all interface objects, all 2017 fitting/validation releases, every fitted object and selection—and was committed and pushed. Uncertainty is a paired household-cluster bootstrap (2,000 replicates over the 10,701 final households) with single-step max- $|t|$  simultaneous intervals inside each pre-declared family.

**Study 2: development refinement.** 2018 only for its registered evidence, with a household-cluster bootstrap over the underlying cohort. The deviation from Study 1 is deliberate and recorded: the 2017 study had one household partition shared by all three seeds and could cluster on it directly, whereas the 2018 pools are split per seed and overlap in people, so clusters are drawn once per replicate from the cohort and applied to every seed. Study 2 later re-scored the 2017 partitions as *exploratory* evidence, refitting probes and attackers on 2017 fitting pools and selecting on 2017 validation pools; it deliberately never calls the sealed loader, and it computes no simultaneous intervals there.

**Data independence.** This is transport between public survey years of the same state and age band, not a fresh sample from the 2018 distribution. Public identifiers *cannot* establish that 2017 respondents are different people from 2018 respondents. Verified disjointness is limited to published household groups within 2017. Income is nominal and unadjusted, and the annual adjustment factor differs between years, which limits cross-year interpretation of the income column. The same caution will apply to any future 2016 evaluation, where the income estimand differs again.

**What was registered in each study, and what was not.** In Study 1, the protocol, the four comparison families, every endpoint, the uncertainty procedure and all decision rules were fixed, hashed and pushed before the final partition could be read—but *no directional prediction was registered*. The result in §7.2 is therefore a prespecified-rule confirmation on a sealed evaluation, not a called shot. It is also not merely exploratory: fixing the contrasts, the endpoint families, the multiplicity correction and the stopping point before the outcome is what makes the simultaneous intervals mean what they say. In Study 2, six directional



paired difference (nats); left of zero favours the candidate. Blue: interval entirely below zero. Orange: entirely above. Grey: crosses zero (unresolved, not unaffected).

**Figure 1:** Paired differences with simultaneous 95% intervals inside each pre-declared family, Mode B (fresh 2017 attacks), both weightings. Each difference is candidate minus comparator in nats. **Left panel, F1:** C1 against its two local controls. The two grey A/sex rows are the unresolved cells—the comparison there is undecided, *not* the endpoint unaffected. **Right panel, F3:** C1 against the frozen neural channel J; every sensitive endpoint is orange.

predictions *were* registered and hashed before fitting; three of the six turned out wrong or ambiguous, and §8.2 reports them before anything else.

**The residence task, stated precisely.** Residence labels were excluded from every representation fit, so no arm’s map saw them. That is the sense in which residence is a held-out task. It is *not* a pristine unseen-task selection: residence outcomes have been used repeatedly across this line of work as the utility endpoint, so the choice of residence as the capability measure is itself a development decision. And exact output preservation does not preserve future service accuracy under shift (§7.7).

## 7 Study 1: the locked 2017 evaluation

### 7.1 Residence capability, and its limits

Every augmented interface clears the 0.01-nat residence reference over H in every seed, weighting and mode; the spectral arms carry 0.025–0.030 nats against 0.017–0.019 for the frozen neural channel J. Table 8 adds the qualification that matters: the stronger half-headroom reference—is the channel’s residence probe within half the gap between the published service and an *unrestricted* PCA32 or richer bank—passes in only 0–2 of 3 seeds for every interface in both modes. The channel adds real capability relative to what is already published. It does not approach what an unrestricted representation of the same covariates would give.

### 7.2 Coalition conditioning beats both local controls, with two cells unresolved

On the sealed 2017 partition, C1 has a simultaneous-interval advantage over both its equal-strength local control L1 and the equal-total-mass control L2, under both weightings. The registered **advantage** rule fired in all four contrast  $\times$  weighting cells, and the registered primary coordination decision was recorded as supported.

The effect is real and it is not uniform. Of the 16 sensitive cells in F1, **14 have adjusted intervals entirely below zero, none is above zero, and two are unresolved:** C1 versus L1 on A/sex unweighted,  $[-0.00585, +0.00020]$ , and C1 versus L2 on A/sex unweighted,  $[-0.00568, +0.00052]$ . In both, the point estimate favours C1 and the same endpoint is strictly better under person weighting. So the accurate statement is that coalition conditioning lowers measured recovery against both local controls, most clearly on

| Contrast                                                                                                                 | Endpoint       | Estimate | Seeds 0 / 1 / 2             | Adjusted 95%           |
|--------------------------------------------------------------------------------------------------------------------------|----------------|----------|-----------------------------|------------------------|
| <i>F1 (registered primary): coalition penalty vs local controls, <math>c = 2.930</math> over 20 endpoints</i>            |                |          |                             |                        |
| C1–L1                                                                                                                    | residence loss | +0.00029 | +0.0005 / +0.0004 / −0.0001 | [−0.00118, +0.00176]   |
| C1–L1                                                                                                                    | A / sex        | −0.00282 | −0.0046 / −0.0055 / +0.0017 | [−0.00585, +0.00020]   |
| C1–L1                                                                                                                    | AB / sex       | −0.00593 | −0.0046 / −0.0094 / −0.0038 | [−0.00915, −0.00271] ▼ |
| C1–L1                                                                                                                    | A / race       | −0.01369 | −0.0145 / −0.0129 / −0.0137 | [−0.01737, −0.01002] ▼ |
| C1–L1                                                                                                                    | AB / race      | −0.00792 | −0.0054 / −0.0109 / −0.0075 | [−0.01105, −0.00478] ▼ |
| C1–L2                                                                                                                    | residence loss | +0.00059 | +0.0012 / +0.0009 / −0.0004 | [−0.00073, +0.00191]   |
| C1–L2                                                                                                                    | A / sex        | −0.00258 | −0.0020 / −0.0051 / −0.0007 | [−0.00568, +0.00052]   |
| C1–L2                                                                                                                    | AB / sex       | −0.00472 | −0.0058 / −0.0060 / −0.0024 | [−0.00804, −0.00140] ▼ |
| C1–L2                                                                                                                    | A / race       | −0.00741 | −0.0054 / −0.0074 / −0.0095 | [−0.01088, −0.00394] ▼ |
| C1–L2                                                                                                                    | AB / race      | −0.00302 | +0.0004 / −0.0039 / −0.0056 | [−0.00524, −0.00081] ▼ |
| <i>F3 (registered secondary): coalition channel vs frozen neural channel J, <math>c = 2.728</math> over 10 endpoints</i> |                |          |                             |                        |
| C1–J                                                                                                                     | residence loss | −0.00932 | −0.0141 / −0.0020 / −0.0118 | [−0.01500, −0.00363] ▼ |
| C1–J                                                                                                                     | A / sex        | +0.01301 | +0.0201 / +0.0034 / +0.0155 | [+0.00753, +0.01848] ▲ |
| C1–J                                                                                                                     | AB / sex       | +0.00697 | +0.0088 / +0.0014 / +0.0107 | [+0.00151, +0.01242] ▲ |
| C1–J                                                                                                                     | A / race       | +0.03767 | +0.0428 / +0.0383 / +0.0320 | [+0.02941, +0.04594] ▲ |
| C1–J                                                                                                                     | AB / race      | +0.03268 | +0.0330 / +0.0318 / +0.0332 | [+0.02504, +0.04032] ▲ |

**Table 3:** Registered families F1 and F3, Mode B, scope `transport_all`, budget 360, unweighted rows shown; the person-weighted rows are in Figure 1. ▼ marks an adjusted interval entirely below zero (candidate better), ▲ entirely above (worse), blank means the interval crosses zero.

| Family                                  | Sens. cells | Better | Worse | Unresolved | advantage fired | $c$   |
|-----------------------------------------|-------------|--------|-------|------------|-----------------|-------|
| F1 primary (C1 vs L1, L2; Mode B)       | 16          | 14     | 0     | 2          | 4 / 4           | 2.930 |
| F2 replication (J vs L025, L20; Mode B) | 16          | 6      | 0     | 10         | 4 / 4           | 3.024 |
| F3 secondary (C1 vs J; Mode B)          | 8           | 0      | 8     | 0          | 0 / 2           | 2.728 |
| F4 frozen transfer (all four; Mode A)   | 32          | 21     | 2     | 9          | 6 / 8           | 3.206 |

**Table 4: The denominator, made explicit.** “Sens. cells” counts contrasts  $\times$  4 sensitive roles  $\times$  2 weightings. “advantage fired” counts the registered rule—at least one sensitive endpoint strictly better and none strictly worse, *per weighting*—over contrast  $\times$  weighting cells. The two are different objects and this paper quotes whichever one it means.

the coalition endpoints and local race, with local sex unweighted undecided. Our earlier summary said “8 of 8 sensitive endpoints better, 0 worse under both weightings”; that count is the endpoint  $\times$  comparator grid scored as better in *at least one* weighting, and as written it overstates the result. It is withdrawn (§9, item 11). Frozen 2018 attacks reproduce the ordering (F4), so it is not an artifact of refitting attackers on the new year.

### 7.3 The residence rule is a point criterion, and only a point criterion

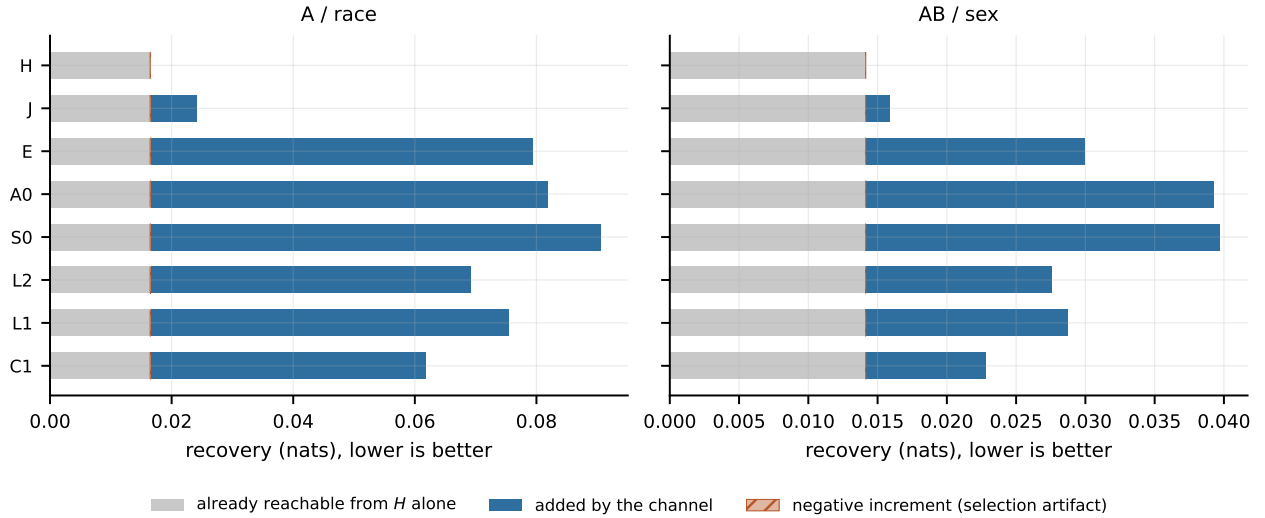
The rule as implemented is  $\text{seed-mean}(\text{C1} - \text{comparator}) \leq 0.001$  on the residence log loss. It is one-sided—C1 may be arbitrarily better—and it is a *point* criterion: no interval enters it. The four values are 0.00029, 0.00010, 0.00059, 0.00017, so it passes, and would pass under a two-sided reading too.

It does not follow that the residence cost is below 0.001. Table 5 shows that **zero of four comparisons support non-inferiority at that margin**, missing it by 2% to 35%, and zero support two-sided equivalence. “No statistically detected cost” and “a cost bounded below 0.001” are different statements and only the first is supported. Our earlier phrasing “inside the prespecified 0.001 band” invited exactly the interval reading the evidence does not support, and we withdraw it.

The same distinction applies to the harm side of the rule. The **advantage** rule requires that *no* sensitive endpoint is significantly worse. That is an absence-of-evidence condition, not a demonstration of no harm: on the two unresolved A/sex cells, up to  $5 \times 10^{-4}$  nats of *extra* local sex recovery is not excluded. We therefore

| Contrast | Weight | Point est. | Pointwise 95% up. | NI?       | Simult. 95% up. | NI?       |
|----------|--------|------------|-------------------|-----------|-----------------|-----------|
| C1 - L1  | unw.   | +0.00029   | +0.00112          | <b>no</b> | +0.00166        | <b>no</b> |
| C1 - L1  | PWGTP  | +0.00010   | +0.00103          | <b>no</b> | +0.00164        | <b>no</b> |
| C1 - L2  | unw.   | +0.00059   | +0.00135          | <b>no</b> | +0.00182        | <b>no</b> |
| C1 - L2  | PWGTP  | +0.00017   | +0.00102          | <b>no</b> | +0.00162        | <b>no</b> |

**Table 5:** Retrospective non-inferiority for the F1 residence endpoint at the registered margin of 0.001 nats. Lower loss is better, so the question is whether the *upper* bound falls below +0.001. It does not, in any of the four comparisons, even under the most permissive procedure (a pointwise one-sided bootstrap percentile with no multiplicity adjustment). This assessment is retrospective and does not replace the registered outcome. The frozen-transfer family F4 is in Appendix B, where 5 of 8 comparisons do clear the margin pointwise and 1 of 8 does so simultaneously.



**Figure 2:** Absolute recovery on the 2017 final partition, split into the part an attacker already gets from the published service alone (grey) and the part the appended channel adds (blue). Mode B, unweighted, seed mean. Lower is better. The grey segment is identical across interfaces by construction: it is H’s own recovery. Hatched segments are *negative* increments, where the validation-selected attack on the augmented wire does worse than H’s own attack.

show point estimates and intervals rather than saying an endpoint is unaffected.

## 7.4 Absolute against additional recovery

The H baseline is substantial: an attacker on the bare service wire already recovers 0.0165 nats of A/race and 0.0141 of AB/sex. Two consequences.

*Ratios are scale-dependent; differences are not.* C1 against J on A/race is  $6.0\times$  on increments (0.0453 vs 0.0076) and  $2.6\times$  on absolute values (0.0617 vs 0.0241). The paired difference, 0.0377 [0.0294, 0.0459], is identical on both scales, because the shared prior and H selection cancel inside a paired contrast. We quote the difference.

*Some increments are negative, and that is a measurement, not protection.* 22 of 429 unweighted role-cells are negative in the primary scope. J’s own A/sex increment is negative under PWGTP ( $-0.0033$ , interval excluding zero) and its absolute A/sex recovery, 0.0092, is *below* H’s own 0.0101: the validation-selected fresh attack on J’s wire does worse than H’s attack. That is a property of selection. H’s attack, routed onto J’s augmented wire, remains executable, so a negative increment removes nothing an H-only attacker already had. In 8 of 312 sensitive cells the selected attack generalises worse than that routed H control (15 of 312 under the fresh-only scope). Adding frozen 2018 candidates lowers additional recovery for every interface because it strengthens the subtracted baseline, not because the wire discloses less. **No finite attack collection establishes an upper bound on what is recoverable;** every number here is a floor.

| Interface | Residence gain | A / race |            | AB / sex |            |
|-----------|----------------|----------|------------|----------|------------|
|           |                | absolute | additional | absolute | additional |
| H         | 0.0000         | 0.0165   | +0.0000    | 0.0141   | +0.0000    |
| J         | 0.0189         | 0.0241   | +0.0076    | 0.0159   | +0.0017    |
| E         | 0.0263         | 0.0794   | +0.0629    | 0.0299   | +0.0158    |
| A0        | 0.0289         | 0.0818   | +0.0653    | 0.0393   | +0.0251    |
| S0        | 0.0304         | 0.0905   | +0.0740    | 0.0397   | +0.0256    |
| L1        | 0.0285         | 0.0754   | +0.0590    | 0.0288   | +0.0146    |
| L2        | 0.0288         | 0.0692   | +0.0527    | 0.0275   | +0.0134    |
| C1        | 0.0282         | 0.0617   | +0.0453    | 0.0228   | +0.0087    |

**Table 6:** Absolute and additional (H-relative) recovery beside residence gain. Mode B, `transport_all`, 360, unweighted, seed mean. Lower is better for recovery, higher for residence gain. The two recovery columns are different quantities and are never combined.

| Comparator  | gain at $p=1$ | $p^*$ | A/race at $p^*$ | reachable? |
|-------------|---------------|-------|-----------------|------------|
| E           | 0.0263        | 1.072 | –               | <b>no</b>  |
| A0          | 0.0289        | 0.976 | 0.0638          | yes        |
| L025        | 0.0175        | 1.615 | –               | <b>no</b>  |
| L20         | 0.0181        | 1.555 | –               | <b>no</b>  |
| J           | 0.0189        | 1.493 | –               | <b>no</b>  |
| S0          | 0.0304        | 0.929 | 0.0688          | yes        |
| C1 (target) | 0.0282        | –     | 0.0453          | –          |

**Table 7:** Post-hoc average-utility matching, Mode B, unweighted.  $p^* = \text{gain}_{C1} / \text{gain}_{\text{source}}$  is the withholding probability at which a comparator’s *mean* residence gain equals C1’s; it is identified in closed form because the mixture is exactly linear in  $p$ . Values above 1 mean the comparator cannot reach C1’s utility at any schedule, and the corresponding recovery cell is therefore undefined rather than blank-because-worse.

## 7.5 The mechanism is not competitive with the frozen neural channel

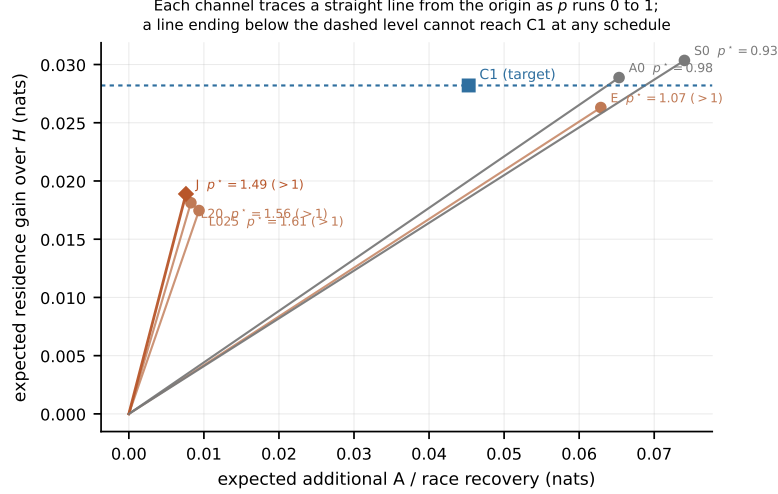
C1 is significantly worse than J on *all four* sensitive endpoints under *both* weightings, and worse on both source probes (income +0.0081, employment +0.0135, intervals excluding zero). It is better only on residence,  $-0.0093 [-0.0150, -0.0036]$  (Figure 1, right panel). The registered secondary decision F3 is therefore recorded as not supported, and its local-race failure clause also fires.

Two things this does *not* say. It does not say J dominates: J carries materially less residence capability than every spectral arm, so neither channel Pareto-dominates the other and we make no such claim. And it does not say the mechanism is behind the state of the art, because J is an internal channel and no external method was run (§5). The honest statement is the narrow one: at matched width, under identical attacks, the spectral channel discloses significantly more of both protected attributes to both views than this particular adversarially trained channel, while supplying more of the authorised capability. “Same width” is a *capacity* control and says nothing about whether their utility is comparable. It is not, which is why §7.6 matches them on it.

## 7.6 Matching the two channels on average utility

Because the routed family mixes per-person *losses*, expected residence gain over H is exactly  $p$  times the full-release gain, and expected additional recovery is exactly  $p$  times the full-release increment. The probability matching a target’s *mean* residence gain is therefore identified in closed form. This yields two findings the fixed- $p$  grid did not surface.

First, **J cannot be made as useful as C1 by any withholding schedule**: it would need  $p^* = 1.49$ . Nor can E (1.07), L025 (1.62) or L20 (1.56). The C1-versus-J comparison is not two points on one traversable frontier. Second, **the ordering survives matching in both directions**. Where matching is identifiable,



**Figure 3:** Post-hoc average-utility matching. Under the declared randomised-withholding mechanism (H always released; an independent visible Bernoulli( $p$ ) branch, fixed per person, decides whether the channel is also released), expected residence gain and expected additional recovery are both exactly  $p$  times their full-release values, so each channel traces a straight line from the origin. Lower and further right is better. The dashed line marks C1’s full-release residence gain; a comparator whose line ends below it cannot reach C1’s utility at any schedule, and its  $p^*$  is printed as  $> 1$  rather than hidden.

C1 beats withheld A0 ( $p^* = 0.976$ ) and withheld S0 ( $p^* = 0.929$ ) on all four sensitive endpoints; and C1 withheld down to J’s residence level ( $p^* = 0.670$ ) still leaks far more than J on all four (0.0303 against 0.0076 on A/race).

Both statements are **post-hoc average-utility matching**:  $p^*$  is estimated from the same data as the outcome, so the fixed- $p$  intervals do not transfer to it, and it matches an average over people who receive different information. The original fixed- $p$  analysis is unchanged, and on it neither family robustly dominates: withholding dominates a given spectral arm in 1–3 of 180 fixed comparisons and a spectral arm dominates withholding in 0–3 of 180. A finite list of controls failing to dominate C1 is not evidence that C1 is efficient.

## 7.7 Service parity, probe quality, and what moved

Three different objects must be kept apart. The *released* income, employment and coverage probability vectors are identical, bitwise, in every condition—structural, verified on 210 of 210 released views. Service *accuracy* changes with the year: employment +0.0138 nats, income −0.0145, coverage −0.0067. Independently fitted *source probes* are a third thing again, and their pass pattern separates the two modes (Table 8).

## 7.8 Development against transport: magnitude, not only resolution

All 20 F1 *seed-mean* signs agree with development. Only 13 of 20 agree in *every seed* (Figure 4); across all 90 family endpoints only 48 of 90 have all three seeds agreeing with their own seed mean. And magnitudes moved: F1 point estimates changed by factors between 0.054× and 14.8× (for instance C1 – L2 on AB/sex went from −0.0003 to −0.0047). A larger evaluation pool shrinks standard errors; it does not move point estimates by a factor of fifteen. The years differ in sample size, in calendar year, in evaluation pool and in which attacker families were available, and this design does not separate them. Our earlier claim that “what changed is resolution, not direction” is withdrawn.

## 7.9 Support limitations

Race class 3 (Alaska Native alone) has one person in the final partition and none in 2017 attacker validation, so class-specific recall and AUROC are undefined for it. Full nine-class log loss is always scored with a fixed  $10^{-12}$

| Interface | source-probe allowance |                | half-headroom residence |        |
|-----------|------------------------|----------------|-------------------------|--------|
|           | Mode A (frozen)        | Mode B (fresh) | Mode A                  | Mode B |
| H         | 3 / 3                  | 3 / 3          | 0 / 3                   | 0 / 3  |
| E         | 3 / 3                  | 3 / 3          | 0 / 3                   | 0 / 3  |
| A0        | 3 / 3                  | 3 / 3          | 0 / 3                   | 1 / 3  |
| J         | 3 / 3                  | 3 / 3          | 0 / 3                   | 0 / 3  |
| S0        | 0 / 3                  | 3 / 3          | 2 / 3                   | 1 / 3  |
| L1        | 0 / 3                  | 3 / 3          | 0 / 3                   | 1 / 3  |
| L2        | 0 / 3                  | 3 / 3          | 0 / 3                   | 1 / 3  |
| C1        | 0 / 3                  | 3 / 3          | 0 / 3                   | 1 / 3  |

**Table 8:** Passing seeds out of 3 for the two reference criteria, unweighted. The legacy source-probe allowance passes everywhere under fresh adaptation (Mode B) but **passes in only 0–1 of 3 seeds for every spectral arm under frozen transfer** (Mode A): a frozen 2018 probe does not carry onto a spectral wire in a new year, and refitting the probe repairs it. Separately, the released service vectors are identical bits in every condition (210 of 210 views), while service *accuracy* moves between the years by +0.0138 nats on employment, −0.0145 on income and −0.0067 on coverage. A worse probe does not change a delivered prediction, and an identical delivered prediction is not a stable one.

| Seed | $q$ | $U$ : pos / neg / $\approx 0$ | L1 pos | L2 pos | C1 pos | $r_+$     |
|------|-----|-------------------------------|--------|--------|--------|-----------|
| 0    | 128 | 32 / 0 / 96                   | 32     | 32     | 32     | <b>16</b> |
| 1    | 128 | 32 / 0 / 96                   | 32     | 32     | 32     | <b>16</b> |
| 2    | 128 | 32 / 0 / 96                   | 32     | 32     | 32     | <b>16</b> |

**Table 9:** Outcome-free rank diagnostic, computed from already-saved 2018 arm spectra before any new fit.  $r_+ = \min(16, \#\{\lambda_i(U - P) > \tau\})$  under the existing relative tolerance convention. No ACS outcome, reserved label, attack or fit is involved.

probability floor and renormalisation; no category is dropped. A comparable-category diagnostic restricted to classes supported in every fitting pool of both modes reproduces the F1 race differences almost exactly (C1 – L1 on A/race: −0.0137 against −0.0137 on all rows). **A class without support is unmeasured, not protected.**

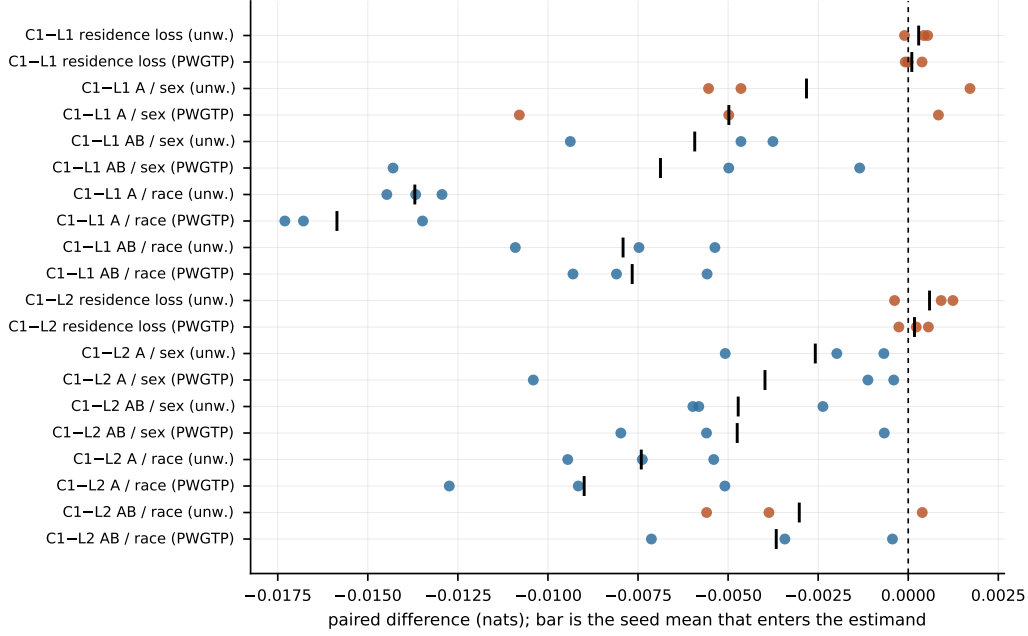
## 8 Study 2: a nonlinear penalty and a rank intervention

Study 2 asked whether the gap in §7.5 is closed by making the penalty act on nonlinear functions of the channel, by reducing the released rank, or by both. Its evidence is ACS 2018 development data throughout; it is not a second confirmation of anything.

### 8.1 The rank question was settled without fitting anything

$r_+ = 16$  in every seed and for every original objective, and not one of the top sixteen directions is nonpositive. Eigenvalue-sign rank selection therefore makes *no change* on this interface: the reduced-rank recipe is an exact alias of the rank-16 recipe, and that column of the design is void by alias. The predeclared branch fired instead, and a rank-8 sensitivity was run for both penalty families and all three policies. **Rank 8 is a compression intervention, not eigenvalue-sign selection**, and it is labelled that way everywhere.

The structural reading is that  $U_{\text{raw}} = (V^\top R/n)(R^\top V/n)$  with  $R$  of width 32, so  $\text{rank}(U) \leq 32$ ; measured,  $U$  has exactly 32 eigenvalues above tolerance and none below, and since the penalties are positive semidefinite,  $\#\text{pos}(U - \lambda P) \leq \#\text{pos}(U)$ . We report the measured counts as the finding. **We do not claim that  $\text{rank}(U) = 32$  by itself implies that all sixteen selected eigenvalues of  $U - P$  are positive**: that needs an inertia argument with assumptions we did not establish. The measurement is what supports the conclusion. And  $r_+$  is the retained rank of the *old fixed linear-moment* objective; the refined objective has no



**Figure 4:** The three per-seed values behind each F1 endpoint; the vertical bar is the seed mean that enters the estimand. Orange marks endpoints where the seeds do not all share the seed-mean sign. Aggregate sign agreement can conceal per-seed disagreement, and does here.

fixed matrix, so neither the rank statement nor Ky Fan optimality transfers to it.

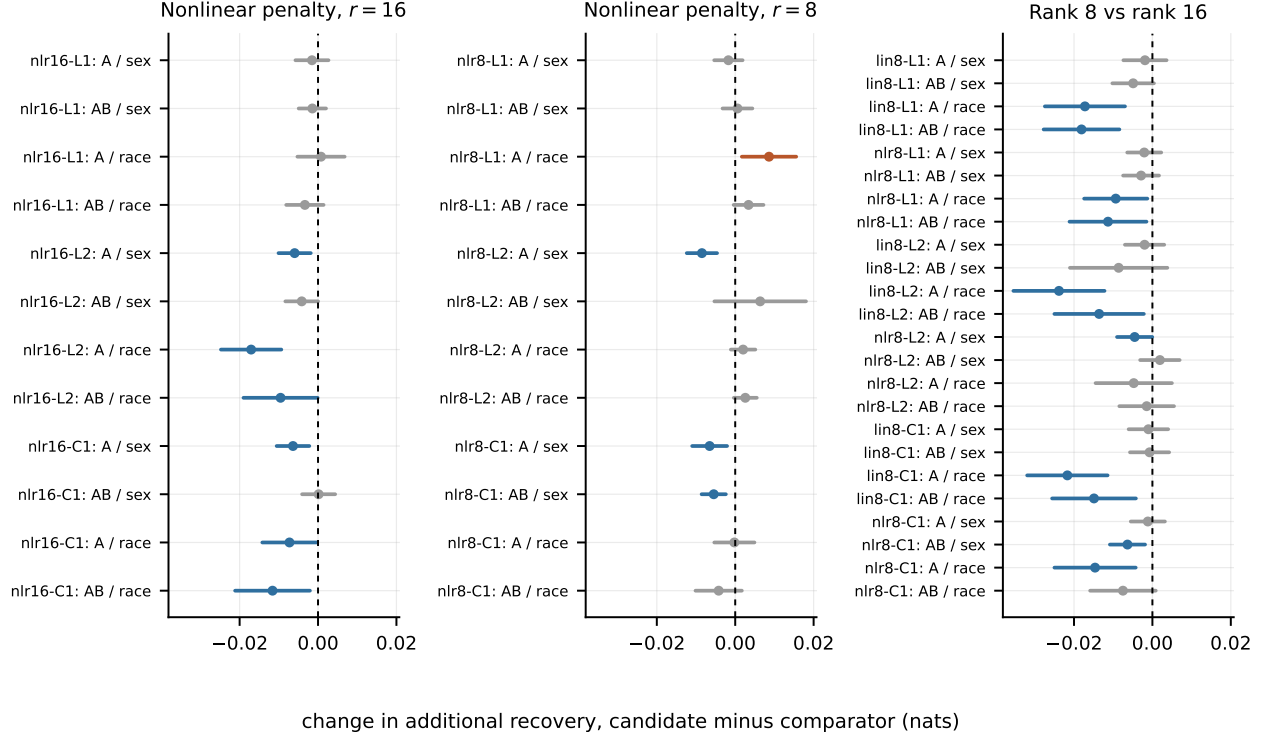
## 8.2 What was registered, and how it turned out

Six directional predictions were registered and hashed before the fit phase started. **Three of the six were wrong or ambiguous.**

- P1  $r_+ = 16$  in all three seeds, so sign-based rank selection changes nothing — **confirmed** (and checkable without a fit).
- P2 The refinement lowers its own training objective below the original solution at rank 16 — **confirmed**, 18/18 conditions, gains 0.017–0.146.
- P3 The best new candidate keeps additional A/race recovery *above* J’s in  $\geq 2$  of 3 seeds — **not confirmed**: 1 of 3 seeds unweighted, 2 of 3 person-weighted; the paired interval on the difference contains zero.
- P4 Training-objective improvement will *not* transfer to independent-attack improvement — **refuted in its qualitative form** (§8.3).
- P5 Rank-8 compression costs  $> 0.002$  nats of residence gain in  $\geq 2$  of 3 seeds — **literally met in 4 of 6 policy×family cells, but the substantive claim fails**: the seed-mean cost is  $\leq 0.002$  everywhere and the sign flips across seeds.
- P6 No new candidate satisfies the primary coordination rule against both local controls — **confirmed**: all eight registered coordination cells are NOT SUPPORTED.

## 8.3 What each factor did, with its denominator

Both factors reduce measured recovery, and they act on different attributes: the nonlinear penalty reduces sex recovery most (A/sex  $-0.0064$  at  $r = 16$ ,  $-0.0065$  at  $r = 8$ ; AB/sex  $-0.0055$  at  $r = 8$ ; also A/race  $-0.0073$  and AB/race  $-0.0116$  at  $r = 16$ ), while rank-8 compression reduces race recovery most (A/race  $-0.0217$  under the original penalty,  $-0.0146$  under the nonlinear one). Composed, they take additional A/race recovery from



**Figure 5:** Each factor separately, unweighted rows, adjusted simultaneous intervals inside each comparison family. Blue: interval entirely below zero (less recovery). Orange: entirely above (more). Grey: crosses zero. The single orange bar in the middle panel is the one cell where the nonlinear penalty made an endpoint significantly worse.

| Factor                     | Held at        | Sens. cells | Better | Worse | Unres. | Residence better/worse |
|----------------------------|----------------|-------------|--------|-------|--------|------------------------|
| Nonlinear penalty          | $r = 16$       | 24          | 7      | 0     | 17     | 0 / 0 (of 6)           |
| Nonlinear penalty          | $r = 8$        | 24          | 5      | 1     | 18     | 0 / 0 (of 6)           |
| Rank-8 compression         | both penalties | 48          | 15     | 0     | 33     | 0 / 0 (of 12)          |
| Coalition vs local penalty | both ranks     | 64          | 29     | 2     | 33     | 0 / 0 (of 16)          |

**Table 10:** The attribution denominators. “Sens. cells” is contrasts  $\times$  4 roles  $\times$  2 weightings. Most cells are unresolved in every factor; the effects are real where they are marked and the majority of the grid is simply undecided at three seeds. The full list of significant cells is Appendix A.

the predecessor’s 0.0274 down to 0.0055, against J’s 0.0071, while holding residence gain at 0.0259 against J’s 0.0215. No residence endpoint moved significantly in either direction in any attribution family.

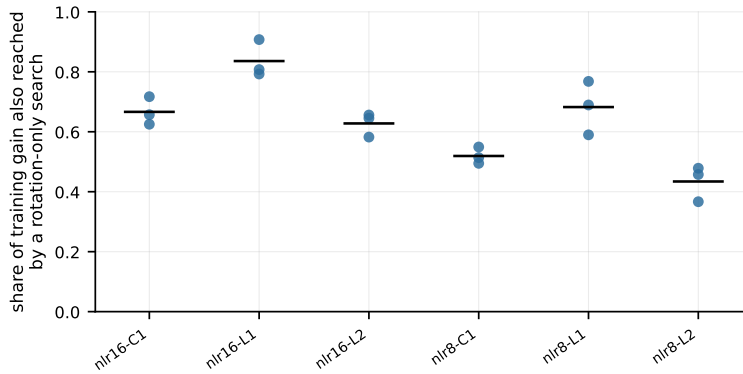
**The denominators matter and were previously misstated.** The nonlinear penalty is significantly better in 7 of 24 sensitive cells at  $r = 16$  and 5 of 24 at  $r = 8$ ; rank-8 compression in 15 of 48; the coalition-versus-local contrast in 29 of 64. And the nonlinear penalty is *not* harmless: at  $r = 8$  with the L1 policy it makes A/race recovery significantly worse, +0.00862 with adjusted simultaneous interval [+0.00168, +0.01556], under unweighted scoring. Study 2’s own decision document says “no endpoint significantly worse” for this factor; that is withdrawn (§9, item 12). Similarly, the coalition penalty is significantly worse than a local control in 2 of 64 cells.

## 8.4 The refined candidate against J: still a no-go

The best new candidate, nlr8-C1, is **significantly worse than J on A/sex** under both weightings (+0.0106 unweighted, +0.0098 person-weighted; adjusted intervals exclude zero). A/sex is a forbidden role and is not set aside. On the other three family sensitive endpoints it is statistically indistinguishable from J, and its

| Endpoint       | Weight | Estimate | Adjusted 95%           |
|----------------|--------|----------|------------------------|
| residence loss | unw.   | −0.00435 | [−0.01238, +0.00368]   |
| A / sex        | unw.   | +0.01065 | [+0.00331, +0.01799] ▲ |
| AB / sex       | unw.   | +0.00247 | [−0.00396, +0.00890]   |
| A / race       | unw.   | −0.00160 | [−0.01160, +0.00840]   |
| AB / race      | unw.   | −0.00170 | [−0.00832, +0.00492]   |
| residence loss | PWGTP  | −0.00432 | [−0.01349, +0.00485]   |
| A / sex        | PWGTP  | +0.01210 | [+0.00441, +0.01978] ▲ |
| AB / sex       | PWGTP  | +0.00422 | [−0.00273, +0.01117]   |
| A / race       | PWGTP  | +0.00133 | [−0.00934, +0.01200]   |
| AB / race      | PWGTP  | +0.00052 | [−0.00713, +0.00818]   |

**Table 11:** The best new candidate, **nlr8-C1**, against the frozen neural channel J on 2018 development data. Two of eight sensitive cells are significantly worse; none is better; the residence advantage is positive in point estimate and not significant.



**Figure 6:** For each condition and seed, the share of the refinement’s training-objective gain that a *separately optimised, information-preserving* orthogonal transformation of the original solution’s subspace also reaches. Bars are condition means. Mean 0.628, range 0.367–0.907 over 18 conditions. The rotation search is itself a bounded optimisation, stopping on an update budget or a failed line search, so each value is an attained share, not a supremum — and it is a property of the *objective*, not an accounting of what the optimiser did.

residence gain is higher (0.0259 against 0.0215) but not significantly so. It also costs more service-probe loss: it clears the legacy source allowance in 1–2 of 3 seeds where J clears it in 3 of 3. All eight registered coordination cells against the local controls fail, reproducing—not resolving—the limitation Study 1 could only settle on a five-times-larger locked partition.

So the decision is: **do not run a confirmation on this design**. That is a decision about this design. It is not evidence that spectral, kernel, nonlinear or fixed-service methods cannot work here; a finite collection of failed variants rules out nothing of the kind. Nor is J’s position an upper bound on attainable utility at a privacy constraint: a maximally leaky channel is not such a bound, and neither is any single measured point.

## 8.5 The rotation diagnostic

The utility term, the original linear penalty and the information content of the release are all invariant under  $W \mapsto WQ$  for orthogonal  $Q$ . The refined penalty is not: measured, the nonlinear penalty changes by up to  $7.5 \times 10^{-3}$  under such a rotation while utility changes by  $\leq 3.3 \times 10^{-16}$  and the original penalty by  $\leq 2.8 \times 10^{-17}$ . So the refined objective is genuinely coordinate-dependent, and a rotation that provably changes nothing an attacker can recover can nonetheless improve it.

The cause is localised exactly and reproduced on a synthetic fixture: summing degree-two monomials  $a \leq b$  with equal weight computes  $\sum_a M_{aa}^2 + \sum_{a < b} M_{ab}^2$  instead of  $\|M\|_F^2$ , and only the latter survives  $M \mapsto Q'MQ$ . Weighting off-diagonals by  $\sqrt{2}$  restores exact invariance ( $< 10^{-12}$  against about 10% slack). Per-feature

standardisation adds about 1%; the Fourier block’s residual slack is Monte-Carlo error that shrinks with feature count.

**What this diagnostic is, and what it is not.** It demonstrates a weakness of the objective: a quantity that should be a function of the released information is not. It is *not* a causal budget partition of the optimiser’s work. We previously wrote that roughly 63% of the optimiser’s progress was “provably inert” and the remaining 37% “bought” the measured privacy reductions, and predicted that removing the inert part would let the same compute buy more. That is withdrawn (§9, item 13). The measured quantity is the share of the training gain *also attainable* by a separately optimised invariant transformation; the two searches are different optimisations over different feasible sets, so their values do not decompose the actual trajectory into an effective part and a wasted part, and nothing here predicts what a repaired objective would achieve. The defect is a reason to fix the construction before spending a confirmation year on it, which is what the next-step specification does; it is not a reason to expect the fix to succeed.

## 8.6 The frozen-nuisance diagnostic

The frozen nuisances barely beat a constant prior on the protected attributes: across the local and coalition roles and all three seeds, sex by 0.03–0.66% of the prior loss and race by 0.96–2.35%. The same nuisance family reaches 8.1–8.9% on public coverage, an *authorised* attribute, so the gap is specific to the protected targets rather than a property of the nuisance machinery. On these fitted models the protected residual is therefore close to the *marginal* residual, which is one reason an  $H$ -conditioned penalty had limited room to differ from the marginal one in this study.

**This is a diagnostic of those frozen nuisance models and nothing more.** It does *not* show that  $H$  carries little population information about sex or race; it does not show that the fitted  $m_j(H)$  are accurate conditional means; and it does not show that conditional and marginal protection coincide in general. A restricted nuisance class can miss information and can miss conditional interactions entirely, and a misspecified  $m(H)$  can manufacture a penalty more than ten times the oracle value under *exact* conditional independence—which one of our own fixtures demonstrates.

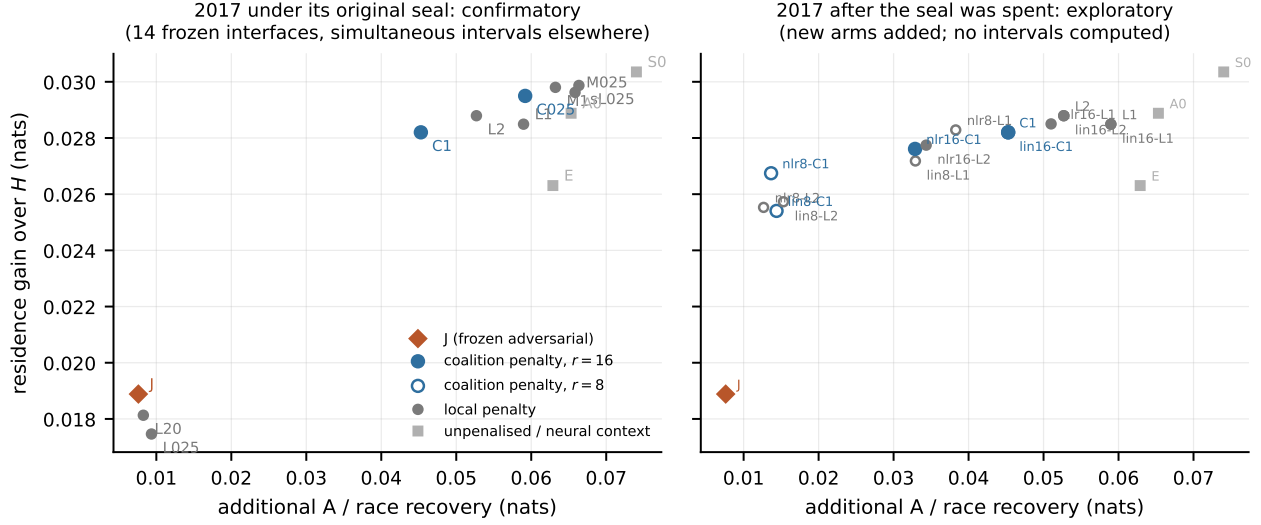
## 8.7 Optimisation quality, stated within its limits

The refinement never returned its initial point in any of the 18 conditions; both restarts completed; and orthogonality held to  $\max |W^\top W - I| \leq 1.4 \times 10^{-15}$ . Those checks rule out three specific failures. **They do not establish that the nonconvex objective was optimised well.** The original fixed-matrix solver has a restricted global optimum by Ky Fan; the refinement has no such guarantee, and a suboptimal solution remains a live explanation for its results.

## 8.8 Exploratory cross-year reuse, which cuts against the candidate

Study 2 also re-scored the 2017 partitions. That evidence is **exploratory**: the 2017 seal was already spent by Study 1, so those rows are now an exposed development pool. Probes and attackers were fitted on 2017 fitting pools and selected on 2017 validation pools, so selection still never saw the evaluation rows, and *no simultaneous intervals are computed there by design*. Years are reported separately and rows are never pooled.

On that partition the two 2018 attribution effects reproduce in direction at the seed mean, and nine conditions reproduce the historical arms digit for digit, confirming the aliasing end to end. But the comparison against J is *worse* here, not better: J has lower additional recovery than **n1r8-C1** on all four family sensitive endpoints (0.0013/0.0039/0.0069/0.0015 against 0.0105/0.0155/0.0127/0.0154), while the candidate keeps its residence advantage (0.0267 against 0.0189). The one place the 2018 result came closest to J—parity on both race endpoints—does not carry over. Per-seed direction agreement with 2018 is only moderate (188 of 288 seed-level comparisons). Figure 7 shows both 2017 roles side by side. We report this because it cuts against the candidate; omitting it would be the selective reading.



**Figure 7:** Residence capability against additional local race recovery. **Left:** the locked 2017 evaluation, the frozen 14-interface slate, confirmatory. **Right:** the same survey year after its seal was spent, with Study 2’s new arms added — exploratory, no intervals. The two panels share axes so the new arms can be placed against the frozen slate, but they are *different evidence*: only the left panel supports a confirmatory reading. In both, J sits alone at low utility and low disclosure.

## 9 Claims this revision withdraws or narrows

Stated explicitly, because each was in one of our own earlier documents. Items 1–10 were already corrected in the previous revision and are repeated here so this paper is self-contained; items 11–16 are new in this integration. Almost all are wording, scope or counting: no numerical result changed, and all 90 Study 1 endpoints reproduce under an independently written scorer to  $1.1 \times 10^{-15}$ .

1. “*Residence difference inside the prespecified  $\pm 0.001$  band.*” The rule is one-sided and applies to a point estimate. Zero of four interval comparisons support non-inferiority at that margin. The registered rule still passed.
2. “*What changed is resolution, not direction.*” Magnitudes changed by  $0.054\times$  to  $14.8\times$ ; sample size alone does not explain the newly significant result.
3. “*Every F1 endpoint keeps its development sign (20/20).*” True of seed means; 13 of 20 in every seed.
4. “*C1 versus J fails on local race.*” It fails on all four sensitive endpoints under both weightings, plus both source probes.
5. “*Several times more sex and race.*” Scale-dependent:  $6.0\times$  on increments,  $2.6\times$  in absolute terms. We now quote the paired difference.
6. “*Under frozen transfer every spectral arm fails in 0–1 of 3 seeds.*” The underlying column counts *passing* seeds; the arms pass in 0–1 of 3, i.e. fail in 2–3 of 3.
7. “*The gap is the surrogate, not the solver.*” Global optimality of the fixed matrix excludes a solver failure and nothing else. Six candidate causes remain unseparated. We now say “consistent with”.
8. “*If a nonlinear penalty also loses, the conclusion generalises to the closed-form spectral family.*” Withdrawn entirely. Two arms failing is evidence about two arms; and a penalty acting on nonlinear functions of  $Z$  is provably not a trace form, so such a successor is not in that family at all.
9. “*Two lock amendments.*” Three exist. All are code-only; the third post-dates every final-partition read.
10. *Reporting the H-relative residence gain without the half-headroom failure.* Corrected in Table 8.

11. “*8 of 8 sensitive endpoints better, 0 worse, under both weightings*” (F1). The denominator is 16 cells, of which 14 are strictly better, 0 worse and 2 unresolved. The registered **advantage** rule—a different object—fired in 4 of 4 cells (§7.2, Table 4).
12. “*No endpoint significantly worse*” for the nonlinear-penalty factor. One is: nlr8-L1 against lin8-L1 on A/race, unweighted, +0.00862 with an adjusted interval excluding zero (§8.3).
13. “*Roughly 37% of the surrogate movement bought the measured reductions, so removing the provably inert 63% should let the same compute buy more.*” Withdrawn as a causal claim and as a prediction. The diagnostic reports an attainable share under a separately optimised invariant transformation, not a decomposition of the optimiser’s trajectory (§8.5).
14. “*Not an optimisation artifact.*” The three completed checks exclude three specific failures and do not establish good optimisation of a nonconvex objective.
15. “*The protected residual is close to the marginal residual, so any H-conditioned penalty has limited room to differ, whatever function class it uses.*” Narrowed to a statement about the frozen nuisance models actually fitted.
16. “*2016’s admission has never been run.*” It has: 2016 is admitted on provenance and schema with a prospective label-blind split prepared, and remains **unscored** (Table 1). The reciprocal stale claim—that the other terminal was still running—is also removed.

## 10 Limitations

Three seeds are three fixed fitted systems on one cohort, not population replications. The bootstraps cover people, not retraining variability and not Census design variance; PWGTP replicate weights were not used. Attack strength is a floor, not a guarantee. Only recipient A receives a channel. Transport is between public survey years of the same state and age band, and cross-year person distinctness cannot be established from public identifiers. Income is nominal and unadjusted, and its estimand differs again in 2016, which any future evaluation there must handle. The evaluation refits attackers but not the representation, so it does not test a mechanism retrained on the new year. The 2018 pools have been used repeatedly and are development, not confirmation; the 2017 seal is spent; Study 2’s evaluated maps carry a known, unrepaired coordinate-dependence defect. No external published method was executed, so no competitiveness claim against the literature is made or supported.

**Numerical recovery.** Three code-only lock amendments are recorded. The first fixed a release-file write race; the second fixed rare load-dependent prediction corruption found by an independent cross-process replay; the third changed three published tables from CSV to gzip CSV and post-dates every final read. Machine memory pressure was the *observed condition* under which the corruption appeared; it is not a demonstrated cause, and we do not claim one. The repair does not depend on the diagnosis: every prediction is computed twice and must agree, all final units were rescored, originals are preserved, the post-fix replay found 0 mismatches in 59,874 predictions with 0 selection changes in 37,122 validation predictions, and Study 2’s independent cross-process replay of 27,540 stored predictions matched bitwise with the machine at swap capacity throughout.

## 11 Conclusion

A service that has already published predictions can add a reusable channel beside them without changing a single released number, and can steer that channel away from directions a recipient or a colluding pair would use to infer sex and race. On a locked, previously unused survey year, the coalition-conditioning knob does what it claims against both of its local controls—14 of 16 sensitive cells strictly better, none worse, two unresolved—under simultaneous intervals and both weightings. It also costs residence utility by an amount the data cannot bound below the 0.001 nats our registered rule used, and the channel it steers remains significantly worse than a frozen adversarially trained channel of the same width on every sensitive endpoint, an ordering that survives matching the two on average utility in both directions. A refinement designed to

close that gap moved measured recovery substantially in the intended direction, failed every one of its eight registered coordination cells, remained significantly worse than the neural channel on local sex, looked worse still on an exploratory cross-year look, and turned out to rest on a coordinate-dependent objective.

The defensible contribution is the problem formulation and the evaluation design: fixed published outputs as a hard constraint, recipient-specific and coalition-specific audit roles, fresh and frozen attackers never pooled, absolute and incremental recovery reported side by side, negative increments retained unclipped, denominators printed beside every count, and a sealed temporal transport with the multiplicity fixed in advance. Both method results are negative and we report them as such.

## 12 Reproducibility

Deterministic seeds throughout; every unit writes a hash-bound completion record; the lock refuses to open the final partition if any hashed input changed. All 90 pre-declared Study 1 endpoints were regenerated from stored per-person predictions by an independently written scorer that shares no computation function with the original report code, agreeing to  $1.1 \times 10^{-15}$  on point estimates and  $2.4 \times 10^{-15}$  on interval endpoints, with identical critical values; candidate selection was re-derived from stored validation losses with 0 mismatches in 924 cells; and the lock was re-verified over all 17,639 hashed inputs with every amendment applied. Study 2’s comparison arithmetic was verified independently from its published CSVs. Every table and figure in this paper is generated by one script from committed evidence files, each recorded with a sha256 in `MANIFEST.json`; no number was typed by hand. Commands, file map and software versions: `REPRODUCE_PAPER.md`. Fitted objects, releases and per-person arrays stay local; compact aggregates, code, tables and figures are published.

## A Significant attribution cells

Every cell in the Study 2 attribution families whose adjusted simultaneous interval excludes zero. ▼: less recovery. ▲: more. Cells not listed are unresolved at three seeds; the denominators are in Table 10.

| Family | Contrast          | Endpoint  | Weight | Estimate   | Adjusted 95%         |
|--------|-------------------|-----------|--------|------------|----------------------|
| C1     | nlr16-L2-lin16-L2 | A / sex   | unw.   | -0.00597 ▼ | [-0.01009, -0.00184] |
| C1     | nlr16-L2-lin16-L2 | A / race  | unw.   | -0.01706 ▼ | [-0.02477, -0.00935] |
| C1     | nlr16-L2-lin16-L2 | AB / race | unw.   | -0.00956 ▼ | [-0.01902, -0.00009] |
| C1     | nlr16-L2-lin16-L2 | A / race  | PWGTP  | -0.01316 ▼ | [-0.02176, -0.00456] |
| C1     | nlr16-C1-lin16-C1 | A / sex   | unw.   | -0.00639 ▼ | [-0.01056, -0.00221] |
| C1     | nlr16-C1-lin16-C1 | A / race  | unw.   | -0.00728 ▼ | [-0.01422, -0.00034] |
| C1     | nlr16-C1-lin16-C1 | AB / race | unw.   | -0.01161 ▼ | [-0.02116, -0.00206] |
| C6     | nlr8-L1-lin8-L1   | A / race  | unw.   | +0.00862 ▲ | [+0.00168, +0.01556] |
| C6     | nlr8-L2-lin8-L2   | A / sex   | unw.   | -0.00850 ▼ | [-0.01237, -0.00463] |
| C6     | nlr8-L2-lin8-L2   | A / sex   | PWGTP  | -0.00634 ▼ | [-0.01054, -0.00214] |
| C6     | nlr8-C1-lin8-C1   | A / sex   | unw.   | -0.00653 ▼ | [-0.01099, -0.00207] |
| C6     | nlr8-C1-lin8-C1   | AB / sex  | unw.   | -0.00548 ▼ | [-0.00858, -0.00238] |
| C6     | nlr8-C1-lin8-C1   | AB / sex  | PWGTP  | -0.00518 ▼ | [-0.00876, -0.00159] |
| C3     | lin8-L1-lin16-L1  | A / race  | unw.   | -0.01721 ▼ | [-0.02744, -0.00698] |
| C3     | lin8-L1-lin16-L1  | AB / race | unw.   | -0.01808 ▼ | [-0.02775, -0.00842] |
| C3     | lin8-L1-lin16-L1  | A / race  | PWGTP  | -0.01336 ▼ | [-0.02493, -0.00178] |
| C3     | lin8-L1-lin16-L1  | AB / race | PWGTP  | -0.01232 ▼ | [-0.02325, -0.00140] |
| C3     | nlr8-L1-nlr16-L1  | A / race  | unw.   | -0.00935 ▼ | [-0.01742, -0.00129] |
| C3     | nlr8-L1-nlr16-L1  | AB / race | unw.   | -0.01133 ▼ | [-0.02115, -0.00151] |
| C3     | lin8-L2-lin16-L2  | A / race  | unw.   | -0.02384 ▼ | [-0.03547, -0.01221] |
| C3     | lin8-L2-lin16-L2  | AB / race | unw.   | -0.01359 ▼ | [-0.02501, -0.00218] |
| C3     | lin8-L2-lin16-L2  | A / race  | PWGTP  | -0.01811 ▼ | [-0.03112, -0.00509] |
| C3     | nlr8-L2-nlr16-L2  | A / sex   | unw.   | -0.00453 ▼ | [-0.00903, -0.00002] |
| C3     | lin8-C1-lin16-C1  | A / race  | unw.   | -0.02169 ▼ | [-0.03196, -0.01143] |
| C3     | lin8-C1-lin16-C1  | AB / race | unw.   | -0.01490 ▼ | [-0.02559, -0.00421] |
| C3     | lin8-C1-lin16-C1  | A / race  | PWGTP  | -0.01667 ▼ | [-0.02864, -0.00470] |
| C3     | nlr8-C1-nlr16-C1  | AB / sex  | unw.   | -0.00636 ▼ | [-0.01085, -0.00187] |
| C3     | nlr8-C1-nlr16-C1  | A / race  | unw.   | -0.01463 ▼ | [-0.02500, -0.00425] |
| C4     | lin16-C1-lin16-L1 | AB / sex  | unw.   | -0.00515 ▼ | [-0.00911, -0.00119] |
| C4     | lin16-C1-lin16-L1 | A / race  | unw.   | -0.01066 ▼ | [-0.01642, -0.00489] |
| C4     | lin16-C1-lin16-L1 | AB / race | unw.   | -0.01427 ▼ | [-0.01980, -0.00874] |
| C4     | lin16-C1-lin16-L1 | AB / sex  | PWGTP  | -0.00603 ▼ | [-0.01069, -0.00137] |
| C4     | lin16-C1-lin16-L1 | A / race  | PWGTP  | -0.01203 ▼ | [-0.01849, -0.00558] |
| C4     | lin16-C1-lin16-L1 | AB / race | PWGTP  | -0.01502 ▼ | [-0.02149, -0.00856] |
| C4     | lin8-C1-lin8-L1   | A / race  | unw.   | -0.01514 ▼ | [-0.02051, -0.00976] |
| C4     | lin8-C1-lin8-L1   | AB / race | unw.   | -0.01108 ▼ | [-0.01564, -0.00653] |
| C4     | lin8-C1-lin8-L1   | A / race  | PWGTP  | -0.01535 ▼ | [-0.02161, -0.00908] |
| C4     | lin8-C1-lin8-L1   | AB / race | PWGTP  | -0.00991 ▼ | [-0.01535, -0.00446] |
| C4     | nlr16-C1-nlr16-L1 | A / sex   | unw.   | -0.00636 ▼ | [-0.01037, -0.00235] |
| C4     | nlr16-C1-nlr16-L1 | A / race  | unw.   | -0.01870 ▼ | [-0.02509, -0.01231] |
| C4     | nlr16-C1-nlr16-L1 | AB / race | unw.   | -0.02253 ▼ | [-0.03232, -0.01273] |
| C4     | nlr16-C1-nlr16-L1 | A / sex   | PWGTP  | -0.00728 ▼ | [-0.01191, -0.00265] |
| C4     | nlr16-C1-nlr16-L1 | A / race  | PWGTP  | -0.01865 ▼ | [-0.02570, -0.01161] |
| C4     | nlr16-C1-nlr16-L1 | AB / race | PWGTP  | -0.01700 ▼ | [-0.02824, -0.00576] |
| C4     | nlr16-C1-nlr16-L2 | AB / sex  | unw.   | +0.00399 ▲ | [+0.00034, +0.00764] |
| C4     | nlr8-C1-nlr8-L1   | A / sex   | unw.   | -0.00544 ▼ | [-0.00884, -0.00204] |
| C4     | nlr8-C1-nlr8-L1   | AB / sex  | unw.   | -0.00702 ▼ | [-0.01050, -0.00353] |
| C4     | nlr8-C1-nlr8-L1   | A / race  | unw.   | -0.02397 ▼ | [-0.03285, -0.01510] |
| C4     | nlr8-C1-nlr8-L1   | AB / race | unw.   | -0.01871 ▼ | [-0.02632, -0.01110] |
| C4     | nlr8-C1-nlr8-L1   | A / sex   | PWGTP  | -0.00595 ▼ | [-0.00977, -0.00212] |
| C4     | nlr8-C1-nlr8-L1   | AB / sex  | PWGTP  | -0.00645 ▼ | [-0.01046, -0.00243] |
| C4     | nlr8-C1-nlr8-L1   | A / race  | PWGTP  | -0.02156 ▼ | [-0.03232, -0.01081] |
| C4     | nlr8-C1-nlr8-L1   | AB / race | PWGTP  | -0.01774 ▼ | [-0.02684, -0.00864] |
| C4     | nlr8-C1-nlr8-L2   | A / sex   | unw.   | +0.00229 ▲ | [+0.00009, +0.00449] |
| C4     | nlr8-C1-nlr8-L2   | AB / sex  | unw.   | -0.00428 ▼ | [-0.00726, -0.00130] |
| C4     | nlr8-C1-nlr8-L2   | A / race  | unw.   | -0.00539 ▼ | [-0.01044, -0.00033] |
| C4     | nlr8-C1-nlr8-L2   | AB / race | unw.   | -0.00868 ▼ | [-0.01433, -0.00302] |
| C4     | nlr8-C1-nlr8-L2   | AB / sex  | PWGTP  | -0.00385 ▼ | [-0.00734, -0.00037] |
| C4     | nlr8-C1-nlr8-L2   | AB / race | PWGTP  | -0.01018 ▼ | [-0.01717, -0.00319] |

## B Frozen-transfer non-inferiority

The same retrospective non-inferiority assessment for family F4 (Mode A, frozen transfer). Unlike F1, 5 of 8 comparisons clear the 0.001 margin under a pointwise one-sided bound and 1 of 8 clears it simultaneously. This is a different family on a different mode and does not transfer to the registered primary result.

| Contrast | Weight | Point est. | Pointwise 95% up. | NI?       | Simult. 95% up. | NI?       |
|----------|--------|------------|-------------------|-----------|-----------------|-----------|
| C1 - L1  | unw.   | +0.00094   | +0.00190          | <b>no</b> | +0.00262        | <b>no</b> |
| C1 - L1  | PWGTP  | +0.00015   | +0.00140          | <b>no</b> | +0.00237        | <b>no</b> |
| C1 - L2  | unw.   | +0.00014   | +0.00084          | yes       | +0.00142        | <b>no</b> |
| C1 - L2  | PWGTP  | -0.00039   | +0.00049          | yes       | +0.00122        | <b>no</b> |
| J - L025 | unw.   | -0.00148   | -0.00030          | yes       | +0.00077        | yes       |
| J - L025 | PWGTP  | -0.00053   | +0.00102          | <b>no</b> | +0.00236        | <b>no</b> |
| J - L20  | unw.   | -0.00127   | +0.00017          | yes       | +0.00131        | <b>no</b> |
| J - L20  | PWGTP  | -0.00104   | +0.00071          | yes       | +0.00219        | <b>no</b> |

## C Design grid and fit counts

Study 2’s grid was frozen before fitting: 2 penalty families  $\times$  2 rank recipes  $\times$  3 policies  $\times$  3 seeds = 54 nominal interface records, of which 27 are unique new fits, 9 are reused historical objects and 18 are void by alias (the  $r_+$  column, §8). A cache read is never counted as a new fit and a resumed unit is counted once. The grid was not expanded in response to a disappointing result. These counts describe implementation effort; they are not scientific evidence and no claim in this paper rests on them.

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